

VR. A Formal System.

A Minimalist Axiomatization of Arithmetic Grounded in Leibnizian Void

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Abstract

We present a formal axiomatic system VR with three primitives $\{\emptyset, \rightarrow, t\}$ and four axioms. The von Neumann natural numbers are constructed as the unfolding of the succession operator t from the initial void $\emptyset$. Equality and distinctness are defined via Leibniz's principle and are not among the primitives. We show that VR is arithmetically equivalent to Peano arithmetic (PA), from which the consistency of VR relative to ZF set theory follows.

Keywords: axiomatic arithmetic, foundations of mathematics, Leibnizian equality, von Neumann ordinals, Peano arithmetic, consistency.

Part I. VR. The Formal System

1. Primitives

Constant: $\emptyset$.

Binary operator: $\rightarrow$ (implication).

Unary operator: t (succession).

2. Axioms

A1. Generativity.

$F \rightarrow F = \top$ and $F \rightarrow \top = \top$, where F is identified with $\emptyset$ at the logical level, and $\top$ is defined as $F \rightarrow F$. From F , via $\rightarrow$, both values $\{F, \top\}$ are reachable. From $\top$, via $\rightarrow$, only $\top$ is reachable.

A2. Implication.

$\rightarrow$ is a function of type $\{F, \top\} \times \{F, \top\} \rightarrow \{F, \top\}$, given by the classical truth table:

$$F \rightarrow F = \top$$

$$F \rightarrow \top = \top$$

$$\top \rightarrow F = F$$

$$\top \rightarrow \top = \top$$

A3. Succession.

The operator t is defined on $\emptyset$ and on every object generated from it. For every x in the domain of t :

$$t(x) = x \cup \{x\}, \text{ so } x \in t(x) \text{ and } x \subset t(x).$$

A4. Induction.

If P is a property of objects of the system, and:

(i) $P(O_0)$ holds,

(ii) for every x : $P(x) \rightarrow P(t(x))$,

then $P(O_n)$ holds for all n .

Equivalent formulation: the O_n exhaust all objects generated from $\emptyset$ via t .

3. Basis

$\{F, \rightarrow\}$ is functionally complete for classical two-valued logic over $\{F, \top\}$.

Derived operators:

$$\neg x := x \rightarrow F$$

$$\begin{aligned}
x \vee y &:= (x \rightarrow y) \rightarrow y \\
x \wedge y &:= \neg(\neg x \vee \neg y) \\
x \leftrightarrow y &:= (x \rightarrow y) \wedge (y \rightarrow x)
\end{aligned}$$

4. Definitions

Def. 1. $\top := F \rightarrow F$.

Def. 2. Leibnizian Equality. $x = y := \forall p(x) \leftrightarrow p(y)$.

Def. 3. Distinctness. $x \neq y := \neg(x = y)$.

Def. 4. The zeroth ordinal: $O_0 := \emptyset$.

Def. 5. Recursively: $O_{n+1} := t(O_n) = O_n \cup \{O_n\}$.

Def. 6. The natural number n is identified with the ordinal O_n .

5. Consequence of A3 and Def. 3

$t(x) \neq x$ for every x in the domain of t .

Proof. The property "contains x as an element" is true for $t(x)$ (since $x \in t(x)$ by A3) and false for x (by the construction of objects via t from $\emptyset$). By Def. 3, $t(x) \neq x$. ■

6. Construction of von Neumann Numbers

$$\begin{aligned}
O_0 &= \emptyset \\
O_1 &= t(O_0) = \{\emptyset\} \\
O_2 &= t(O_1) = \{\emptyset, \{\emptyset\}\} \\
O_3 &= t(O_2) = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\
&\dots
\end{aligned}$$

Each O_n contains all preceding $O_0, \dots, O_{n-1}$ as elements.

7. Arithmetic

Def. 7. Addition.

$$\begin{aligned}
a + O_0 &:= a \\
a + t(b) &:= t(a + b)
\end{aligned}$$

Def. 8. Multiplication.

$$\begin{aligned}
a \times O_0 &:= O_0 \\
a \times t(b) &:= (a \times b) + a
\end{aligned}$$

Def. 9. Exponentiation.

$$\begin{aligned}
a^{O_0} &:= t(O_0) = O_1 \\
a^{t(b)} &:= a^b \times a
\end{aligned}$$

Theorems (derived by induction via A4).

T1. Commutativity of addition: $a + b = b + a$.

T2. Associativity: $(a + b) + c = a + (b + c)$.

T3. Distributivity: $a \times (b + c) = (a \times b) + (a \times c)$.

T4. $O_1 + O_1 = O_2$.

8. Structure of the System

Level	Content
0. Primitives	$\emptyset, \rightarrow, t$
1. Axioms	A1, A2, A3, A4
2. Logic	Classical two-valued (A1–A2)
3. Equality and Distinctness	Leibnizian (derived)
4. Objects	von Neumann ordinals O_n (A3 + A4)
5. Arithmetic	$+, \times, ^$ (A4)

Part II. Equivalence of VR and Peano Arithmetic

In this part we show that VR is arithmetically equivalent to Peano arithmetic (PA). The consistency of VR relative to ZF then follows as a direct consequence of the consistency of PA relative to ZF.

9. Direction VR $\rightarrow$ PA

We show that all five Peano axioms are derivable in VR as theorems.

Peano Axioms (Standard Formulation).

- P1. $0 \in \mathbb{N}$.
- P2. For every $n \in \mathbb{N}$, there exists $S(n) \in \mathbb{N}$.
- P3. For every $n \in \mathbb{N}$: $S(n) \neq 0$.
- P4. For any $n, m \in \mathbb{N}$: if $S(n) = S(m)$, then $n = m$.
- P5. If $K \subseteq \mathbb{N}$, $0 \in K$, and for every $n \in K$: $S(n) \in K$, then $K = \mathbb{N}$.

Translation into the Language of VR.

- $\mathbb{N} \mapsto \{O_n : n \geq 0\}$
- $0 \mapsto O_0$
- $S \mapsto t$

Theorem 1 (P1 in VR). O_0 is an object of the system.

Proof. By Def. 4: $O_0 := \emptyset$. By the statement of primitives of VR, $\emptyset$ is a constant of the system. Therefore O_0 exists. ■

Theorem 2 (P2 in VR). For every O_n , $t(O_n)$ exists, and $t(O_n)$ is an object of the system.

Proof. By A3, the operator t is defined on $\emptyset$ and on every object generated from it. We show by induction (A4) that t is defined on every O_n .

Base case: $O_0 = \emptyset$. By A3, $t(\emptyset) = \emptyset \cup \{\emptyset\}$ exists.

Step: assume $t(O_n)$ is defined. Then $O_{n+1} := t(O_n)$ is an object generated from $\emptyset$ via t . By A3, t is defined on every such object. Hence $t(O_{n+1})$ is defined.

By A4, $t(O_n)$ is defined for all n , and $t(O_n) = O_{n+1}$ by Def. 5. ■

Theorem 3 (P3 in VR). For every n : $t(O_n) \neq O_0$.

Proof. Consider the property $p(x) :=$ "x contains at least one element": $p(x) \iff \exists y: y \in x$.

We have:

- $p(\emptyset) = F$ (the empty set contains no elements),
- $p(t(O_n)) = T$ for every n (since $O_n \in t(O_n)$ by A3).

The property p distinguishes $t(O_n)$ and $O_0 = \emptyset$. By Def. 3, $t(O_n) \neq O_0$. ■

Theorem 4 (P4 in VR). If $t(x) = t(y)$, then $x = y$.

Proof. Let $t(x) = t(y)$. By A3: $x \cup \{x\} = y \cup \{y\}$.

By Def. 2, for every property q : $q(x \cup \{x\}) \leftrightarrow q(y \cup \{y\})$.

Take the property $q_x(w) := "x \in w"$. We have $q_x(x \cup \{x\}) = \top$ (since $x \in \{x\} \subset x \cup \{x\}$). From the equality $x \cup \{x\} = y \cup \{y\}$ it follows that $q_x(y \cup \{y\}) = \top$, that is, $x \in y \cup \{y\}$. By the definition of union: $x = y$ or $x \in y$.

By symmetry, $y \in x \cup \{x\}$, hence $y = x$ or $y \in x$.

We show that the cases $x \in y$ and $y \in x$ are excluded for objects of VR. Chains of membership in objects generated from $\emptyset$ via t are finite and terminate at $\emptyset$. Proof by induction via A4:

Base case: $O_0 = \emptyset$ has no elements.

Step: $O_{n+1} = O_n \cup \{O_n\}$. Elements of O_{n+1} are either elements of O_n (chains finite by induction) or O_n itself (the chain $O_{n+1} \ni O_n \ni \dots \ni \emptyset$ is finite).

Therefore cycles in $\in$ are impossible in VR. Hence $x \in y \wedge y \in x$ is excluded, and $x = y$. ■

Theorem 5 (P5 in VR). The Peano induction axiom.

Proof. A4 literally states: if $P(O_0)$ and for every x : $P(x) \rightarrow P(t(x))$, then $P(O_n)$ for all n . This is precisely P5 in our notation. ■

Corollary.

All Peano axioms (P1–P5) are derivable in VR as theorems. Therefore every theorem of PA that depends on P1–P5 and classical first-order logic has a corresponding proof in VR.

10. Direction $PA \rightarrow VR$

We show that every theorem of VR has a corresponding representation in PA via coding.

Method.

Encoding of VR objects as natural numbers of PA via arithmetization of syntax (Gödel numbering). To each object of VR we assign a natural number:

$$\ulcorner \emptyset \urcorner := 0,$$

$$\ulcorner t(x) \urcorner := \text{a computable function of } \ulcorner x \urcorner.$$

Under a suitable choice of encoding, $\ulcorner O_n \urcorner = n$. The set-theoretic operations ($\emptyset$, $\{\cdot\}$, $\cup$) are encoded as primitive recursive functions on natural numbers. This is standard technique; see Mendelson, Introduction to Mathematical Logic; Boolos & Jeffrey, Computability and Logic.

Translation of VR Axioms into PA.

A1, A2 (logic): derivable in PA, since PA contains classical first-order logic.

A3 ($t(x) = x \cup \{x\}$): expressible as a primitive recursive relation between $\ulcorner x \urcorner$ and $\ulcorner t(x) \urcorner$.

A4 (induction): literally the induction axiom of PA.

Translation of Leibnizian Equality.

Equality $\ulcorner x \urcorner = \ulcorner y \urcorner$ of natural numbers in PA corresponds to equality of objects in VR. The quantifier "for all properties" in Def. 2 is interpreted as a schema over all arithmetical formulas of PA.

11. Equivalence Theorem

From directions 9 and 10 we obtain:

Theorem. VR and PA are arithmetically equivalent: the $\mathbb{N}$ -theoretic content of one system corresponds bijectively to the $\mathbb{N}$ -theoretic content of the other.

12. Consistency Relative to ZF

Peano arithmetic is consistent relative to ZF: the von Neumann model $\mathbb{N} \subseteq \text{ZF}$ realizes all PA axioms as ZF theorems. This is a classical result; see any standard textbook on set theory.

Corollary. VR is consistent relative to ZF.

Proof. VR is arithmetically equivalent to PA (Theorem 11). PA is consistent relative to ZF (classical result). Therefore any contradiction in VR would entail a contradiction in PA, and hence in ZF. If ZF is consistent, so is VR. ■

The chain is established:

$$\text{VR} \leftrightarrow \text{PA} \hookrightarrow \text{ZF}$$

Part III. Open Questions

The established consistency of VR relative to ZF makes the system suitable for extensions. The following remain open:

(1) Extensions of Number Systems.

Construction of integers $\mathbb{Z}$, rationals $\mathbb{Q}$, and reals $\mathbb{R}$ on top of VR. The standard path involves introducing ordered pairs (after Kuratowski), equivalence classes, Dedekind cuts, or Cauchy sequences. This requires extending the language of VR with a pair construction.

(2) Connection with Type Theory.

Formalization in the style of Martin-Löf, where t is the succ constructor, $\emptyset$ is the zero constructor, and $A4$ is the Nat eliminator. Leibnizian equality naturally aligns with path equality in homotopy type theory (HoTT). This would yield a ready categorical semantics: VR as an initial t -algebra.

(3) Liar Paradox and Tarski's Theorem.

If VR is semantically closed, check the applicability of Tarski's undefinability theorem. A possible direction is the construction of a level hierarchy via iteration of t (each application of t creating a new level on which truth of the previous level can be discussed).

(4) Formalization in a Proof Assistant.

Implementation of VR in Coq, Agda, or Lean for machine verification of all derivations and explicit construction of the $VR \rightarrow PA$ chain in a formally verifiable form.

Summary

VR is a formal axiomatization with three primitives $\{\emptyset, \rightarrow, t\}$ and four axioms. Equality and distinctness are derived via Leibniz's principle. The von Neumann natural numbers are constructed as the unfolding of succession from the initial void; induction guarantees the completeness of this unfolding. Peano arithmetic is fully derived. Consistency relative to ZF is established via arithmetical equivalence with PA.

The main content of VR is not the novelty of arithmetical results (it is equivalent to PA), but the compactness and minimalism of its axiomatization with a transparent motivation for its primitives: $\emptyset$ as Leibnizian initial givenness, $\rightarrow$ as the minimal logical operator, and t as the generative operator of counting. All standard properties of the natural numbers are obtained from these three primitives and four axioms.

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